Team Out of the Box

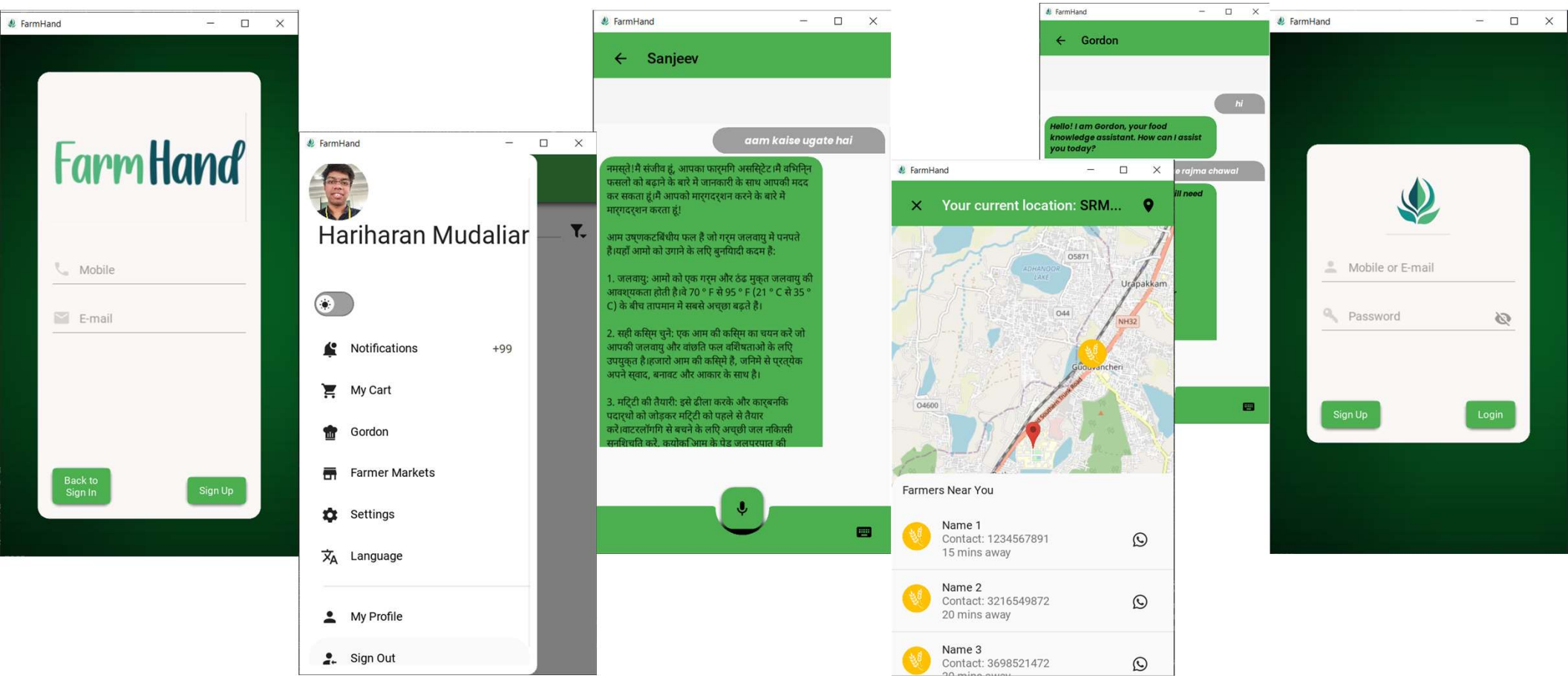
Designation	Name	Register Number	Section
Leader	Hariharan G. Mudaliar	RA2211026010183	Y1
Member	Vansh Vineet Bhatia	RA2211026010197	Z1
Member	Ananya Parasrampuria	RA2211026010299	W2
Member	Debojyoti Mondal	RA2211026010195	Z1

Problem Statements

- To provide Fresh Farm Produce to Consumers in Urban areas by connecting them directly to the nearby Farmers and Farmer Markets based on a Geolocation Sorting system.
- To help Consumers get healthy recipes, ideas, their nutritional values and list of required vegetables, fruits and spices using AI.
- To help Farmers grow better crops by helping them detect the Plant Diseases using their Mobile Camera.
- To eliminate the need of middlemen in the distribution sector.

FarmHand

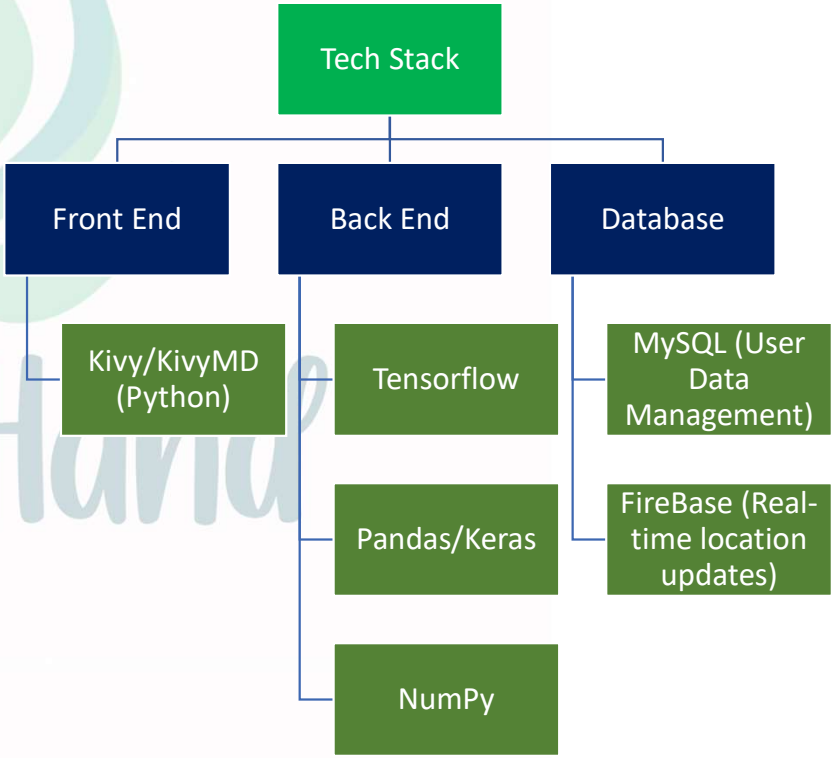
FarmHand App



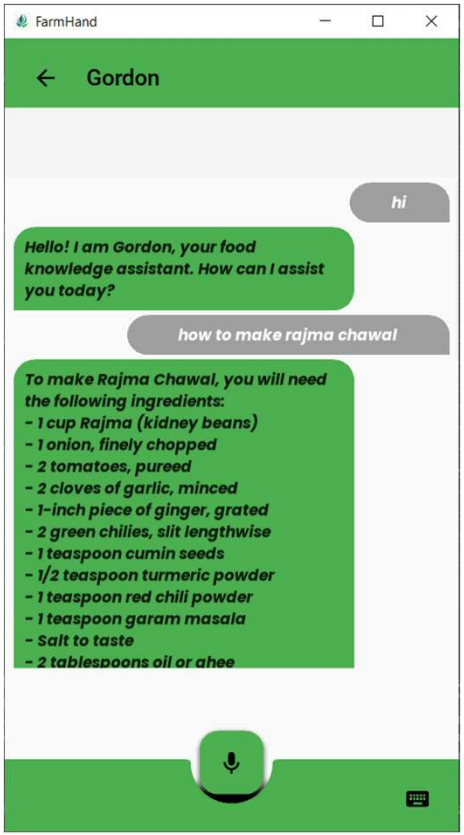
Approach Details

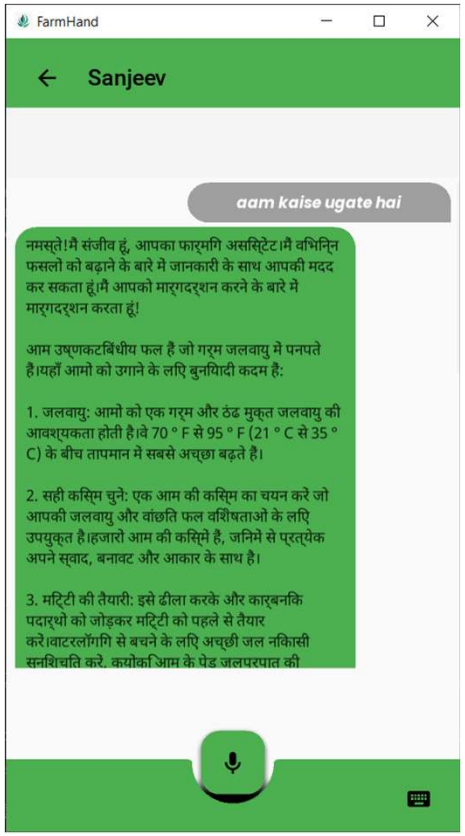
Key Features

- Real-time geolocation-based Farmer tracking system.
- AI Assistant with Multilingual support.
- Plant Health detection using Convolutional Neural Networks.
- Cross-Platform availability.



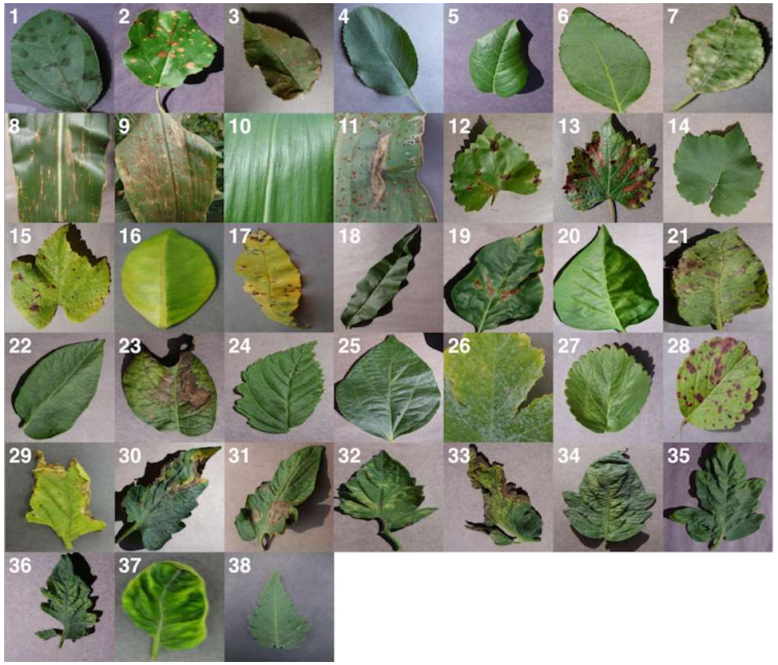
Gordon and Sanjeev





- AI Assistant
- Helps in anything related to food and crops.
- Helps get alternate recipes for seemingly unhealthy food.
- Intuitive UI.
- Multilingual support
- NLP Model:
  - OpenAI davinci-002
- Dataset:
  - https://www.kaggle.com/datasets/kaggle/recipe-ingredients-dataset
  - https://www.sanjeevkapoor.com/AllRecipes
  - https://www.penmai.com/community/threads/chef-damodaran-recipes-vegetarian.69389/
  - https://www.tarladalal.com/all-recipes-308427c
- Libraries:
  - TensorFlow
  - Keras
  - NumPy

Plant Health Detection



- Dataset Description:
  - 54,306 images of plant leaves
  - Spread of 38 class labels
  - PlantVillage dataset (14 crop species with 26 diseases)

Resources:

https://www.plantvillage.org/en/plant_images
https://github.com/salathegroup/plantvillage_deeplearning_paper_dataset

Model Description

1. Choice of deep learning architecture:

- AlexNet (6 layers)
- GoogLeNet (22 layers)

2. Choice of training mechanism:

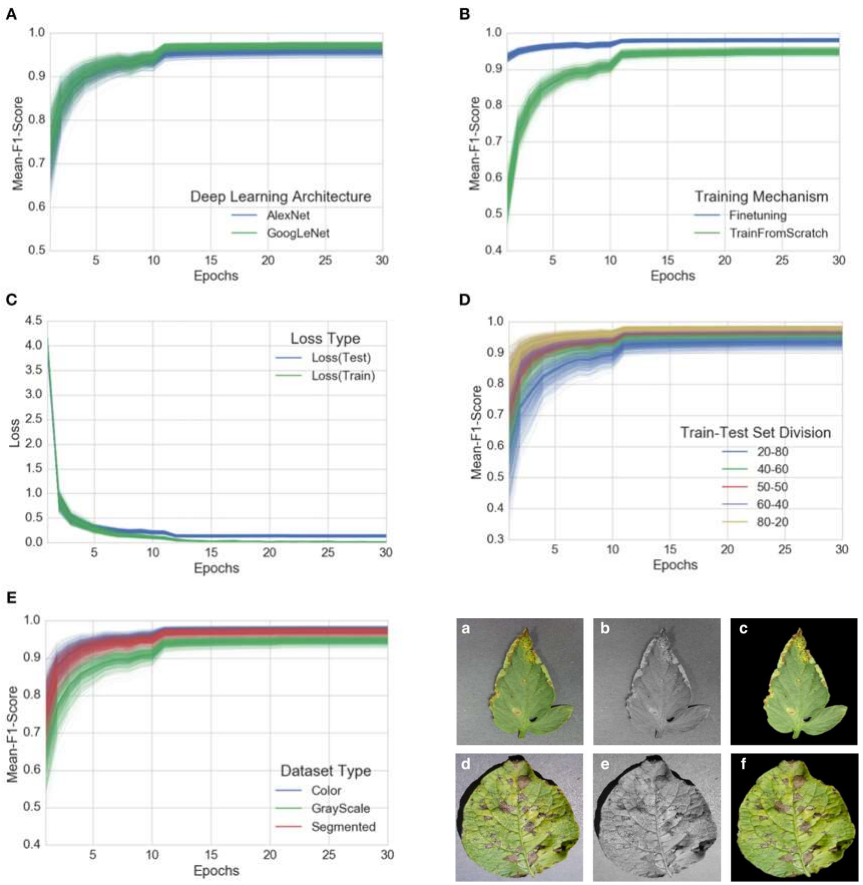
- Transfer Learning,
- Training from Scratch.

3. Choice of dataset type:

- Color,
- Gray scale,
- Leaf Segmented.

4. Choice of training-testing set distribution:

- Train: 80%, Test: 20%,
- Train: 60%, Test: 40%,
- Train: 50%, Test: 50%,
- Train: 40%, Test: 60%,
- Train: 20%, Test: 80%.



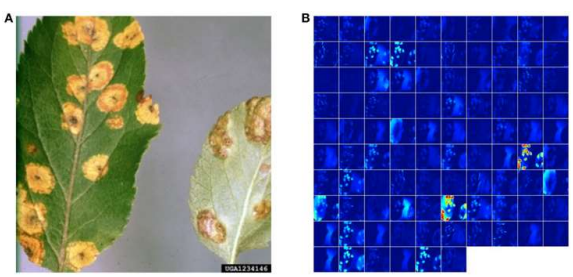
Hyperparameters

- Solver type: Stochastic Gradient Descent,
- Base learning rate: 0.005,
- Learning rate policy: Step (decreases by a factor of 10 every 30/3 epochs),
- Momentum: 0.9,
- Weight decay: 0.0005,
- Gamma: 0.1,
- Batch size: 24 (in case of GoogLeNet), 100 (in case of AlexNet)

Results

	AlexNet		GoogleNet	
	Transfer learning	Training from scratch	Transfer learning	Training from scratch
TRAIN: 200%, TEST: 80%				
Color	0.9736 (0.9742, 0.9737, 0.9738)	0.9118 (0.9137, 0.9132, 0.9130)	**0.9820** (0.9824, 0.9821, 0.9821)	0.9430 (0.9440, 0.9431, 0.9429)
Grayscale	0.9361 (0.9368, 0.9369, 0.9371)	0.8524 (0.8539, 0.8555, 0.8553)	**0.9563** (0.9570, 0.9564, 0.9564)	0.8828 (0.8842, 0.6835, 0.8841)
Segmented	0.9724 (0.9727, 0.9727, 0.9726)	0.8945 (0.8956, 0.8963, 0.8969)	**0.9808** (0.9810, 0.9808, 0.9808)	0.9377 (0.9388, 0.9380, 0.9380)
TRAIN: 400%, TEST: 60%				
Color	0.9860 (0.9861, 0.9861, 0.9860)	0.9555 (0.9557, 0.9558, 0.9558)	**0.9914** (0.9914, 0.9914, 0.9914)	0.9729 (0.9731, 0.9729, 0.9729)
Grayscale	0.9584 (0.9588, 0.9589, 0.9588)	0.9088 (0.9090, 0.9101, 0.9100)	**0.9714** (0.9717, 0.9716, 0.9716)	0.9361 (0.9364, 0.9363, 0.9364)
Segmented	0.9812 (0.9814, 0.9813, 0.9813)	0.9404 (0.9409, 0.9408, 0.9408)	**0.9896** (0.9896, 0.9896, 0.9898)	0.9643 (0.9647, 0.9642, 0.9642)
TRAIN: 50%, TEST: 50%				
Color	0.9896 (0.9897, 0.9896, 0.9897)	0.9644 (0.9647, 0.9647, 0.9647)	**0.9916** (0.9916, 0.9916, 0.9916)	0.9772 (0.9774, 0.9773, 0.9773)
Grayscale	0.9661 (0.9663, 0.9663, 0.9663)	0.9312 (0.9315, 0.9318, 0.9319)	**0.9788** (0.9789, 0.9788, 0.9788)	0.9507 (0.9510, 0.9507, 0.9509)
Segmented	0.9867 (0.9868, 0.9868, 0.9869)	0.9551 (0.9552, 0.9555, 0.9556)	**0.9909** (0.9910, 0.9910, 0.9910)	0.9720 (0.9721, 0.9721, 0.9722)
TRAIN: 600%, TEST: 40%				
Color	0.9907 (0.9908, 0.9908, 0.9907)	0.9724 (0.9725, 0.9725, 0.9725)	**0.9924** (0.9924, 0.9924, 0.9924)	0.9824 (0.9825, 0.9824, 0.9824)
Grayscale	0.9686 (0.9689, 0.9688, 0.9688)	0.9388 (0.9396, 0.9395, 0.9391)	**0.9785** (0.9789, 0.9786, 0.9787)	0.9547 (0.9554, 0.9548, 0.9551)
Segmented	0.9855 (0.9856, 0.9856, 0.9856)	0.9595 (0.9597, 0.9597, 0.9596)	**0.9905** (0.9906, 0.9906, 0.9906)	0.9740 (0.9743, 0.9740, 0.9745)
TRAIN: 80%, TEST: 20%				
Color	**0.9927** (0.9928, 0.9927, 0.9928)	**0.9782** (0.9786, 0.9782, 0.9782)	**0.9934** (0.9935, 0.9935, 0.9935)	**0.9836** (0.9839, 0.9837, 0.9837)
Grayscale	**0.9726** (0.9728.0.9727, 0.9725)	**0.9449** (0.9451, 0.9454, 0.9452)	**0.9800** (0.9804, 0.9801, 0.9798)	**0.9621** (0.9624, 0.9621, 0.9621)
Segmented	**0.9891** (0.9893, 0.9891, 0.9892)	**0.9722** (0.9725, 0.9724, 0.9723)	**0.9925** (0.9925, 0.9925, 0.9924)	**0.9824** (0.9827, 0.9824, 0.9822)

Each cell in the table represents the mean F_1 score$_{(mean\ precision,\ mean\ recall,\ overall\ accuracy)}$ for the corresponding experimental configuration.
The bold values are the F_1 scores of the best performing models in the respective row/column.

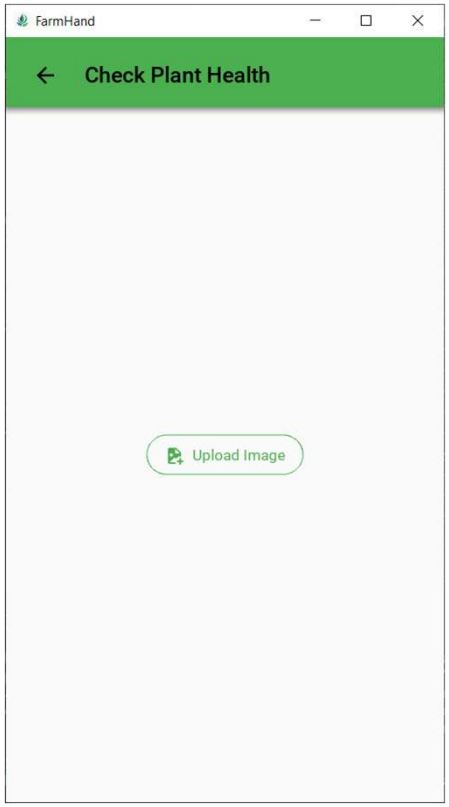


Overall Accuracy:

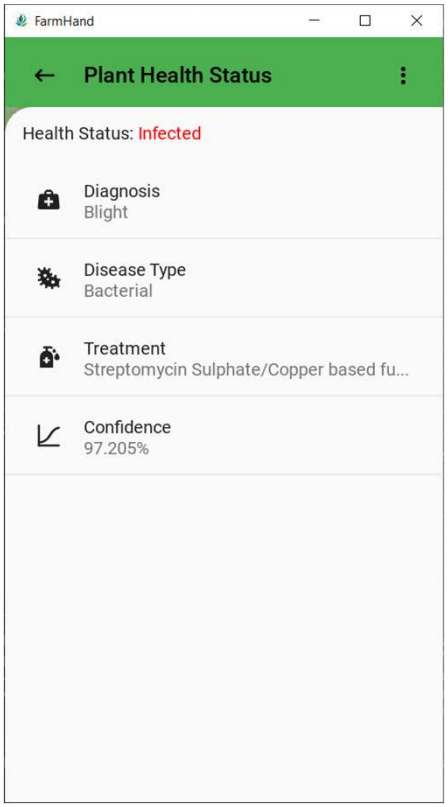
85.53% (in case of *AlexNet::TrainingFromScratch::GrayScale::80–20*)

to

99.34% (in case of *GoogLeNet::TransferLearning::Color::80–20*)


FarmHand
Check Plant Health
Upload Image

FarmHand
Plant Health Status
Health Status: Infected


FarmHand
Plant Health Status
Health Status: Infected
Diagnosis
Blight
Disease Type
Bacterial
Treatment
Streptomycin Sulphate/Copper based fu...
Confidence
97.205%

FarmHand’s not just a Project, it’s a dream…

A dream towards a Stronger, Smarter, Safer and Healthier Future

Thank You